

Oral Antifungals

Oral antifungals are reserved for widespread or more inflammatory lesions. Comparative studies in adults show that fluconazole, 150 mg weekly for 4 to 6 weeks; itraconazole, 100 mg daily for 15 days; and terbinafine, 250 mg daily for 2 weeks are as effective as griseofulvin, 500 mg daily for 2 to 6 weeks, with no significant differences in adverse events. Safe and effective regimens for children include ultramicrosize griseofulvin, 10 to 20 mg/kg/day for up to 6 weeks; itraconazole, 5 mg/kg/day for 1 week; and terbinafine, 3 to 6 mg/kg/day for 2 weeks.

Tinea Pedis

Mild interdigital tinea pedis without bacterial involvement can be treated topically with an allylamine, azole, ciclopirox, benzylamine, tolnaftate, or undecenoic acid. Topical terbinafine for 1 week is 66% effective.

The newer oral antifungals have replaced griseofulvin as the treatments of choice for severe or refractory tinea pedis, especially when accompanied by onychomycosis. The recommended dosing schedule for terbinafine is 250 mg daily for 2 weeks. Effective regimens for itraconazole in adults include 200 mg twice daily for 1 week, 200 mg daily for 3 weeks, or 100 mg daily for 4 weeks, whereas children should receive 5 mg/kg/day for 2 weeks. Fluconazole, 150 mg weekly for 3 to 4 weeks or 50 mg daily for 30 days, is also effective. Associated onychomycosis is common; if present, treatment of the nail infection is necessary to prevent recurrence of tinea pedis.

Maceration, denudation, pruritus, and malodor warrant evaluation for bacterial co-infection via Gram stain and culture. Antibiotics should be initiated once bacterial infection is confirmed and selected based on sensitivity results.

Vesiculobullous tinea pedis results from a T-cell-mediated immune reaction; therefore, symptomatic relief with topical or systemic corticosteroids may be appropriate at the onset of antifungal therapy.

Onychomycosis

Treatment of onychomycosis depends on the severity of nail involvement and the causative organism. Onychomycosis can be categorized based on whether the matrix is involved. In cases without matrix involvement, topical therapy alone may be sufficient. However, oral or combination therapy is recommended for cases with matrix involvement.

Ciclopirox (8% nail lacquer) is a topical agent that can be effective for tinea unguium, applied daily for 48 weeks. In mild to moderate disease, it achieves mycologic cure and clear nails in approximately 7% of cases. It also has activity against *Candida* species and some molds. Despite its lower efficacy compared to oral antifungals, ciclopirox avoids risks of systemic drug interactions, making it a consideration for long-term use in older patients.

Amorolfine is another topical agent formulated as a nail lacquer and belongs to the morpholine derivative class. It is effective against dermatophytes, yeasts, and molds and can be applied once weekly.

Oral antifungals are indicated for refractory, severe, or nondermatophyte onychomycosis, or when a shorter treatment duration or higher cure rate is desired. Selection of an agent should be guided by the causative organism, potential adverse effects, and risk of drug interactions.

Terbinafine is fungicidal against dermatophytes, *Aspergillus*, and *Scopulariopsis*, but has variable activity against *Candida* species.